Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble

Recall at top 10%

NLL per mutation

A

All substitutions

PSSM (sequence composition) 0.88
antiGen (multi−tree) 0.86
antiGen (single−tree) 0.83
Where x which 0.79
Frequency (memorization) 0.79
Substitution matrix 0.70
Where only 0.50
Codon accessibility 0.32
Uniform 0.10

0.00 0.25 0.50 0.75
Recall at top 10%

B

antiGen (multi−tree) 2.40
antiGen (single−tree) 2.23
Where x which 1.77
PSSM (sequence composition) 1.67
Frequency (memorization) 1.55
Substitution matrix 1.42
Codon accessibility 0.77
Where only 0.41
Uniform 0.00

0.0 0.5 1.0 1.5 2.0 2.5
Nats better than uniform

C

De novo only

PSSM (sequence composition) 0.07
antiGen (multi−tree) 0.48
antiGen (single−tree) 0.35
Where x which 0.50
Frequency (memorization) 0.06
Substitution matrix 0.50
Where only 0.15
Codon accessibility 0.36
Uniform 0.10

0.0 0.1 0.2 0.3 0.4 0.5
Recall at top 10%

D

antiGen (multi−tree) 0.11
antiGen (single−tree) −0.60
Where x which 0.93
PSSM (sequence composition) −0.70
Frequency (memorization) −0.19
Substitution matrix 0.97
Codon accessibility 0.88
Where only 0.01
Uniform 0.00

−0.5 0.0 0.5 1.0
Nats better than uniform

All substitutions · De novo only · 95% interval excludes zero · interval includes zero

E

Versus single−tree

SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%

F

SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

−4 −2 0
Multi−tree minus comparator, NLL per mutation

G

Versus where x which

SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

−0.50 −0.25 0.00 0.25
Multi−tree minus comparator, recall at top 10%

H

SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

−2 0 2 4 6 8
Multi−tree minus comparator, NLL per mutation

Pooled−validation epoch (deployed) · Per−tree best epoch

I

Epoch rule

NLL per mutation · *Recall at top 10%*

single−tree model, All substitutions
single−tree model, De novo only
where x which, All substitutions
where x which, De novo only

−1.0 −0.5 0.0 0.5 1.0 | 0.0 0.1
Pooled estimate

J

NLL per mutation · *Recall at top 10%*

single−tree model, All substitutions
single−tree model, De novo only
where x which, All substitutions
where x which, De novo only

−1.0 −0.5 0.0 0.5 1.0 | 0.0 0.1 0.2
Pooled estimate